hala tuju = direction

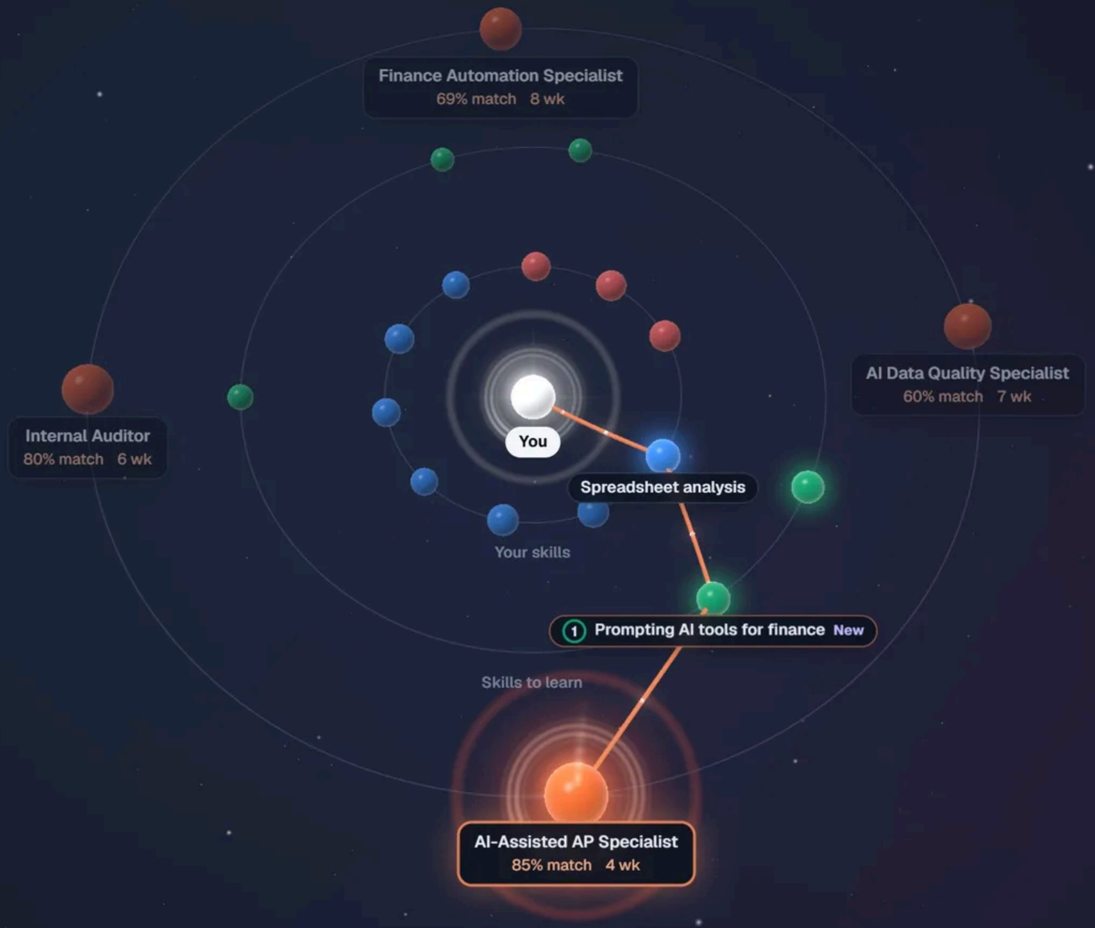
HalaTuju

A GPS for your career: when AI closes a road, it reroutes you.

SDC Hackathon 2026, Domain 1: Aiversecity Future-Ready Workforce

TEAM DSAISE

Leow Shen En
Lim Jie Shin
Choong Zhuo Lin
Tan Wei Feng
Esther Kong Yuan Er



The problem

AI is reshaping jobs one task at a time

620k

Malaysian jobs highly impacted by AI and digitalisation within 3–5 years

TalentCorp impact study, 2024

1 in 4

workers worldwide have a job that AI can partly do. Most of these jobs will change, not vanish.

ILO Working Paper 140, 2025

57%

of AI use at work helps a person do the task. The other 43% does it alone.

Anthropic Economic Index, 2025

“

For many jobs, AI can only do 30–40 percent of the work now and for the foreseeable future. That means we still need people to do the remaining 60–70 percent.

Andrew Ng, World Economic Forum, Davos, January 2026

> HalaTuju shows each worker their 60–70%: the skills AI can’t do, and the jobs those skills lead to.

The gap

Today's tools match skills, not routes

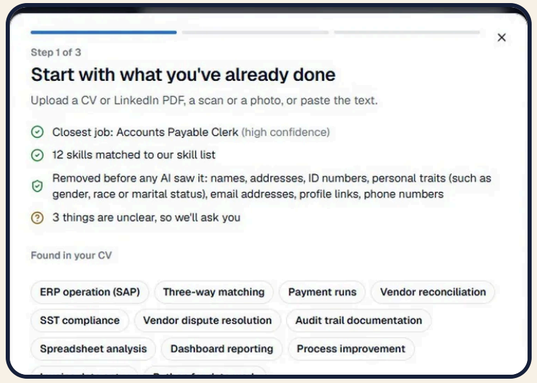
	Skills matching	AI exposure	Step-by-step plan with dates	Updates when AI changes
MyMahir TalentCorp	Skills test and role comparison	Job-level impact study	Upskilling suggestions	*Not stated*
MYFutureJobs PERKESO	AI matching to vacancies	*Not stated*	Reskilling programmes	*Not stated*
Career Explorer LinkedIn	Skills similarity score	*Not stated*	Skill gaps and courses	*Not stated*
HalaTuju this project	✓ 13,939 skills, 3,039 jobs	✓ Per task, from 4 studies	✓ Week by week, with costs	✓ Daily, checked by a person

Sources: Bernama 2024, PERKESO 2025–26, LinkedIn 2020. "Not stated" means not in public material as of Oct 2026.

The product

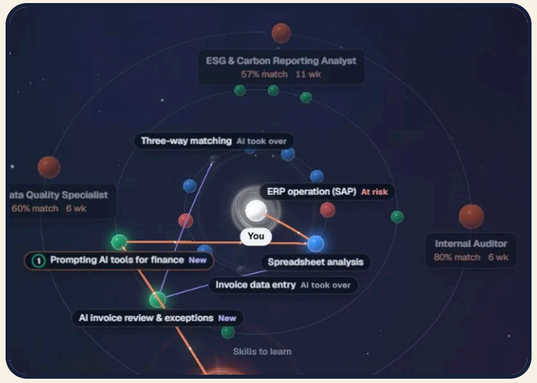
How HalaTuju helps a worker

1 Upload your CV



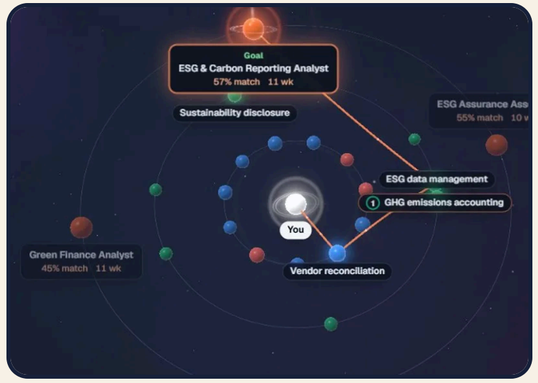
Your private details are removed first. Your skills appear in seconds.

2 See your skills



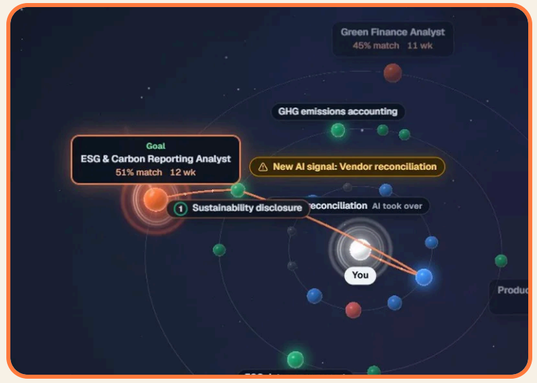
See which skills will still count, and which AI is taking over.

3 Pick a route



Choose a new job and get a study plan, with time, cost and courses.

4 Stay on track

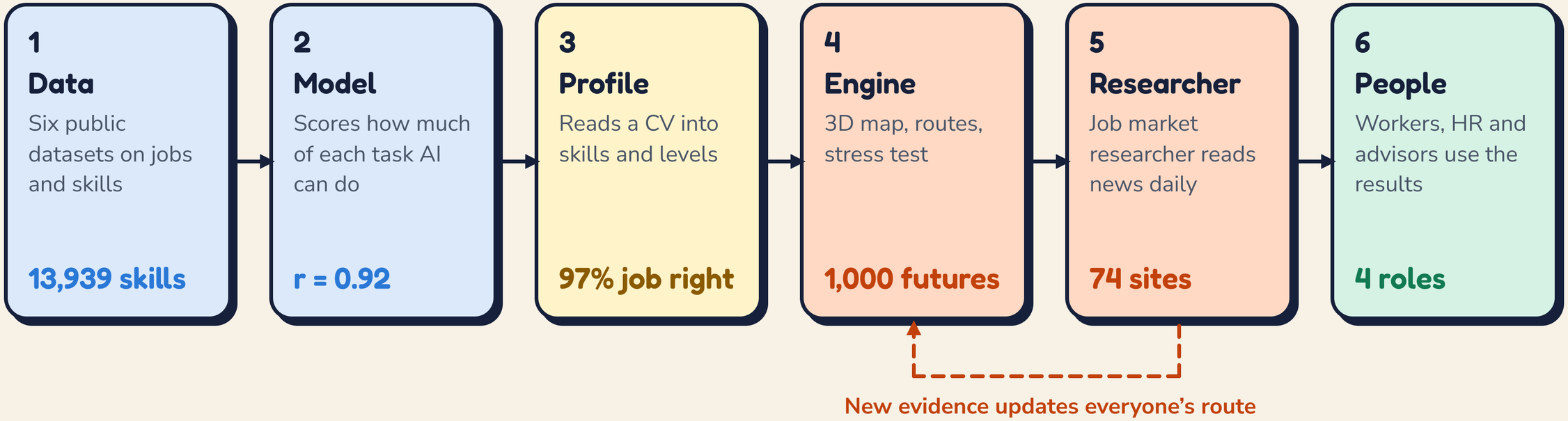


Our job market data stays up to date, and your plan adjusts as it changes.

Screens from the live app at halatuju.replit.app.

The system

How it’s built: six layers, from data to people



Runs on Replit with TypeScript and Postgres. Gemini Flash-Lite reads, bge-m3 on RunPod matches, and 419 automated tests check it.

Built on six public datasets

13,939
skills across 3,039 jobs

ESCO v1.2.1, European Commission

16,411
tasks on ESCO-linked jobs, rated by experts for AI

O*NET with GPTs-are-GPTs labels

2,956
jobs scored for generative-AI exposure

ILO Working Paper 140, 2025

Real use
how people actually use AI at work

Anthropic Economic Index

2021–25
wages and employment, 9 job groups

DOSM, via OpenDOSM

66
critical occupation groups in Malaysia

TalentCorp MyCOL 2024/25

Plus 30 real course listings with price, hours and link. Every number in the app names its source.

Our AI-exposure model agrees with experts

All 19,265 tasks rated by researchers → bge-m3 turns each task into numbers, on RunPod → A simple model learns the expert ratings → Scores all 13,939 skills

0.92

agreement with experts on 185 jobs it never saw

Better than GPT-4's ratings (0.85) and a simple baseline (0.88)

0.79

agreement with the ILO's index on 2,956 jobs

An independent check: we never trained on ILO data

4

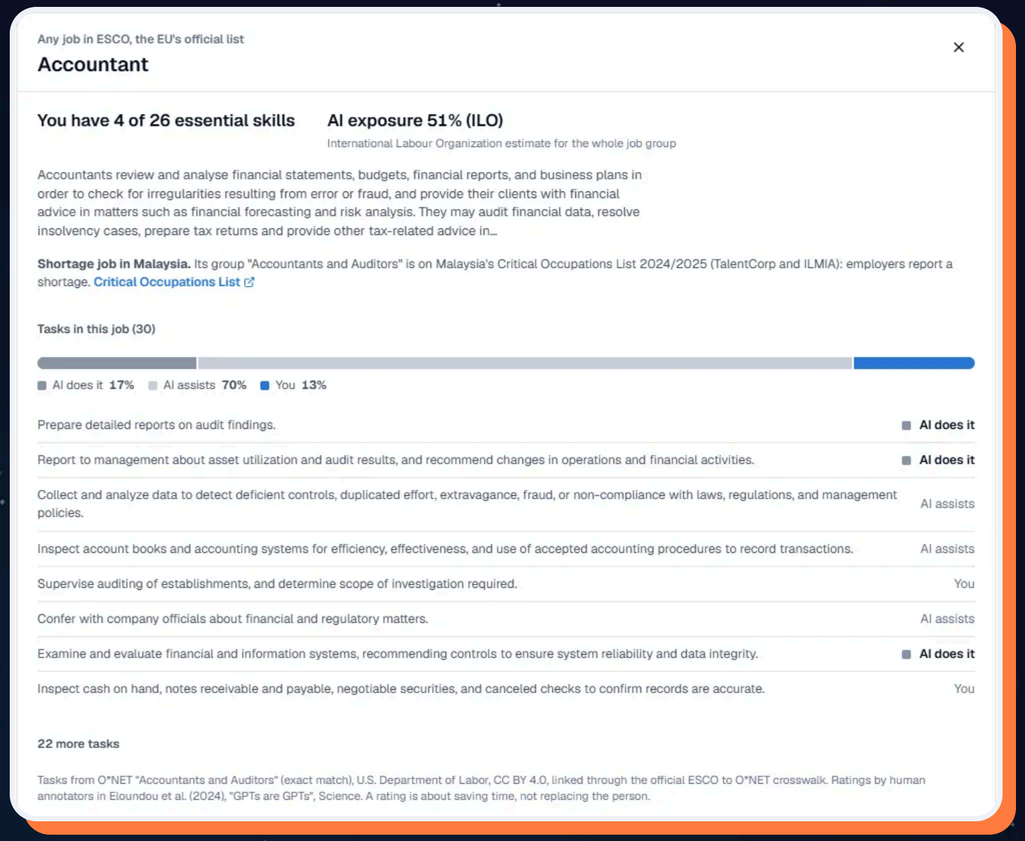
AI-exposure sources shown side by side

ILO, Anthropic, GPTs-are-GPTs and our model

Expert ratings from Eloundou et al., Science 2024. Test jobs were kept out of training. r is the correlation per job (1 = perfect).

1 Data | 2 Model | 3 Profile | 4 Engine | 5 Job market researcher | 6 People

See how AI affects any of 3,039 jobs



Your match

You have 4 of the 26 essential skills

AI exposure from four studies

35–59% for accountants, from the ILO, Anthropic, GPTs-are-GPTs and our model

Task split

Of 30 tasks, AI does 17%, helps with 70%, and 13% still need a person

Malaysian context

On the critical-jobs list, RM 6,124 median pay, 9.8% more workers since 2021

Live screenshot. Sources: ESCO v1.2.1, O*NET via the official crosswalk, ILO WP140, DOSM 2025 and MyCOL 2024/25.

From CV to skill profile in 15 seconds

1 Upload
CV or LinkedIn PDF, or pasted text

2 Scrub
Names, MyKad, phone, religion removed before any AI reads it

3 Match
Gemini reads it; bge-m3 matches ESCO skills

4 Ask
Up to 3 follow-up questions, EN or BM

5 Level
1–5 per skill: from CV, self-rated or quiz-verified

97%
of current jobs identified (35 of 36)

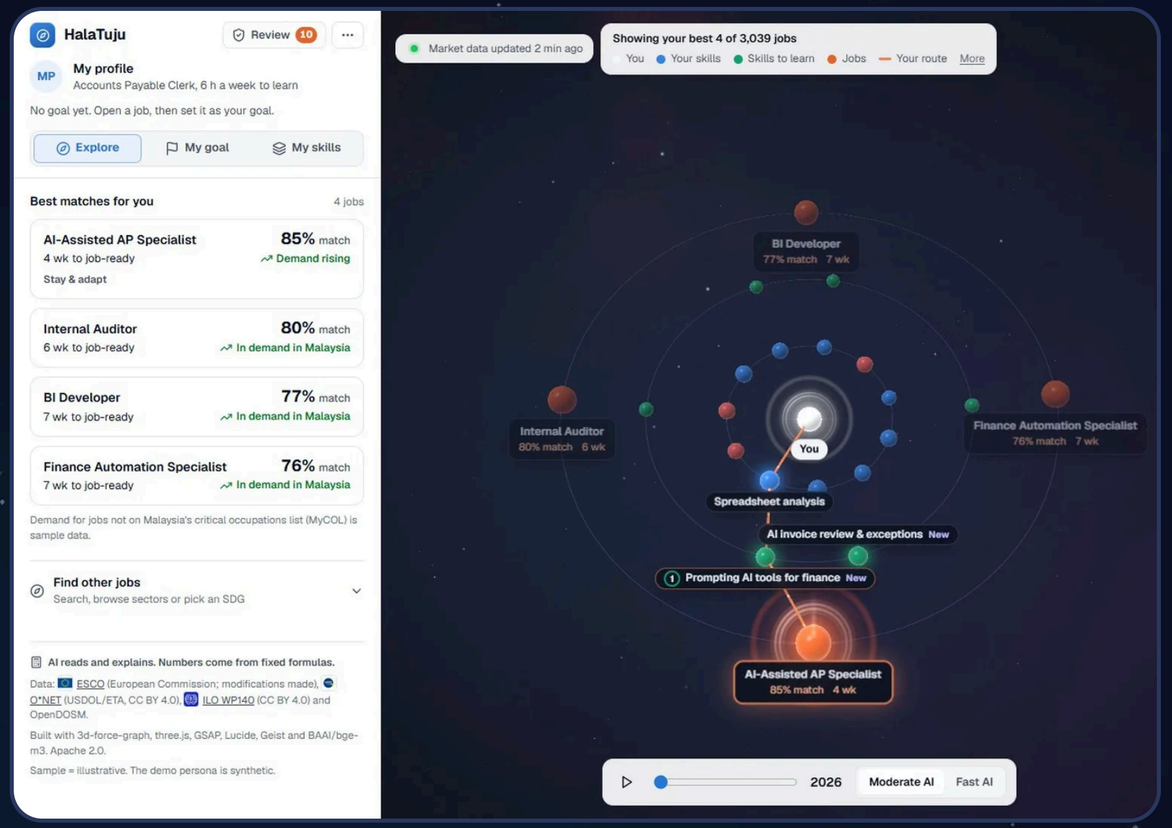
94%
of the expert's map skills found (84 of 89)

15 s
from upload to skill map, Priya's CV

99.4%
of personal details removed before the AI (154 of 155)

Blind test, 24 Oct 2026: 36 synthetic Malaysian CVs we never tuned on, answer key by an independent annotator.

The 3D career map



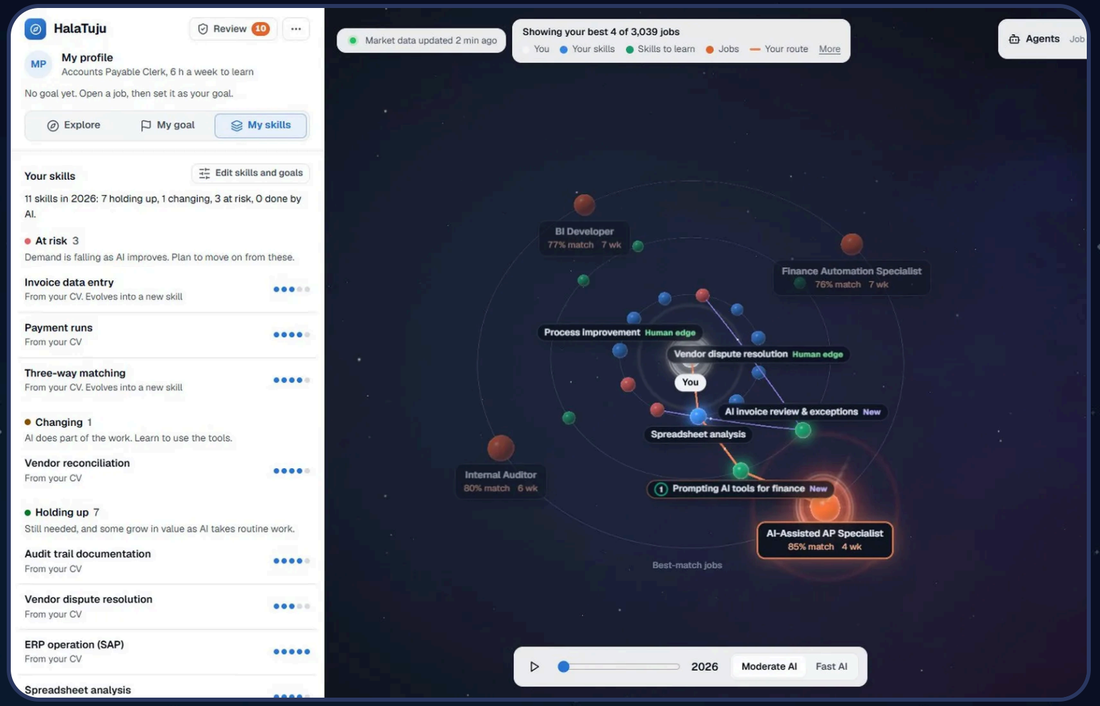
- **Centre: you**
- **Inner ring: your skills.** Red means AI already does it
- **Middle ring: skills to learn,** including new skills
- **Outer ring: jobs,** with match % and weeks
- **Orange line: your route,** step by step

Controls: year 2026–2030, steady or fast AI spread, sector and SDG focus. Click any node for its details.

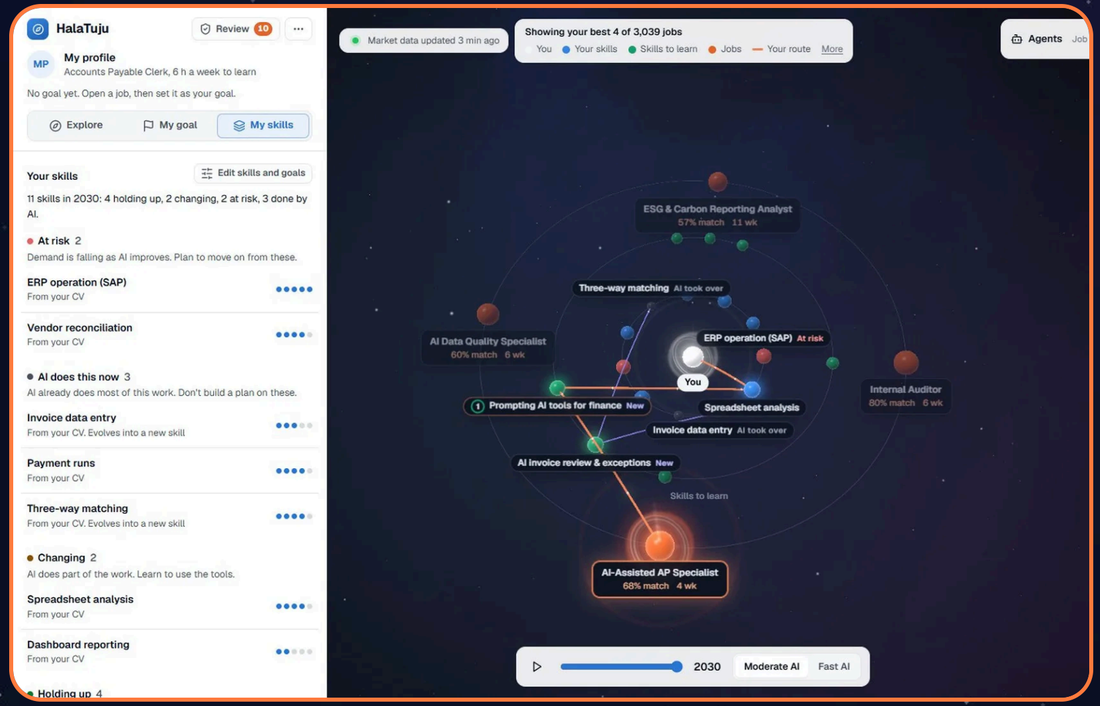
From the demo video: Priya, an Accounts Payable Clerk, and her best 4 of 3,039 jobs. Sample values are labelled in the app.

A time machine: see your skills in 2030

2026: 7 of her 11 skills hold up

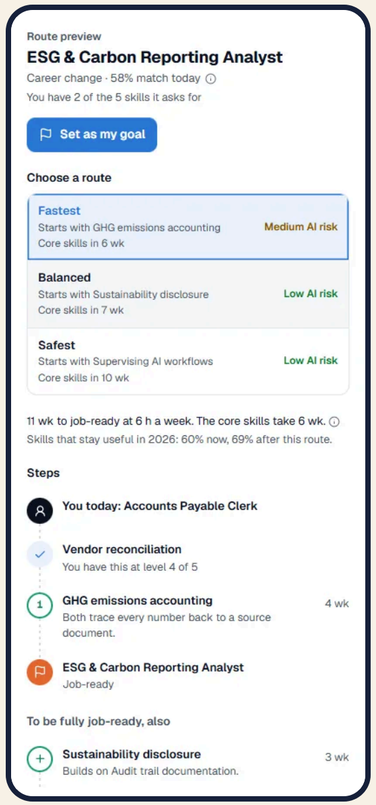


2030: AI does 3 of them; 4 hold up



From the demo video. Invoice entry goes to AI in 2027, three-way matching in 2028. Switch AI spread to Fast and it happens sooner.

Three routes, priced in weeks and RM



Fastest
Core skills in 6 wk
Medium AI risk

Balanced
Core skills in 7 wk
Low AI risk

Safest
Core skills in 10 wk
Low AI risk

How a route is found

Like a GPS, it searches every path from your skills to the job. Each step costs weeks of study at your pace, plus extra for skills AI may take. Safest avoids those most.

Step types

Each step is reuse, upskill, evolve or reskill, done as learn, practice, prove

Costed

Course fees in RM, and how much HRD Corp can cover

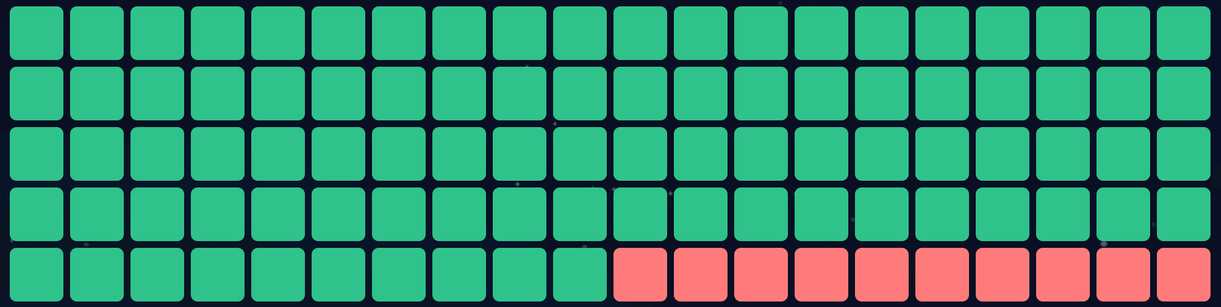
Deadline

Shows if you are ahead, on track or behind your date

Example: AP clerk to ESG analyst in 11 weeks at 6 hours a week, with free courses.

Every plan is tested in 1,000 futures

Priya’s safest route: works in 90% of 1,000 futures



Each square is 10 simulated futures. Red, about 1 in 10: not ready by the deadline. Safest steers around skills AI is taking, so it is the longest route and runs late when study drops below 4.5 hours a week.

Random in each future

- AI speed, from slow to fast
- Our model’s measured error
- Study hours, half to 1.5 times the plan
- Each step’s length, ±20%

Fastest 99%, **Balanced** 100%, **Safest** 90%

The app suggests a fix: take Balanced (100%), or study 1 more hour a week (98%).

Seeded Monte Carlo: same plan, same number, within about ±2 points. It tests a plan; it does not predict hiring.

An AI researcher checks 74 trusted sites daily

An AI agent that reads AI and job news every day, so the skill map and every route stay current.

1 Collect
RSS, arXiv, OpenAlex, GDELT, web search on vetted sites

→

2 Extract
Gemini finds a dated claim about a skill, with a quote

→

3 Verify
The quote must appear word for word on the page

→

4 Review
A second model checks the claim against its evidence

→

5 Triage
Small changes apply; the rest go to a person

769
articles in one run

5.5 min
per run

28
targeted searches

30
AI fact checks

7
flagged for a person

WHAT IT LOOKS FOR

New AI abilities
AI can now do part of a skill's tasks

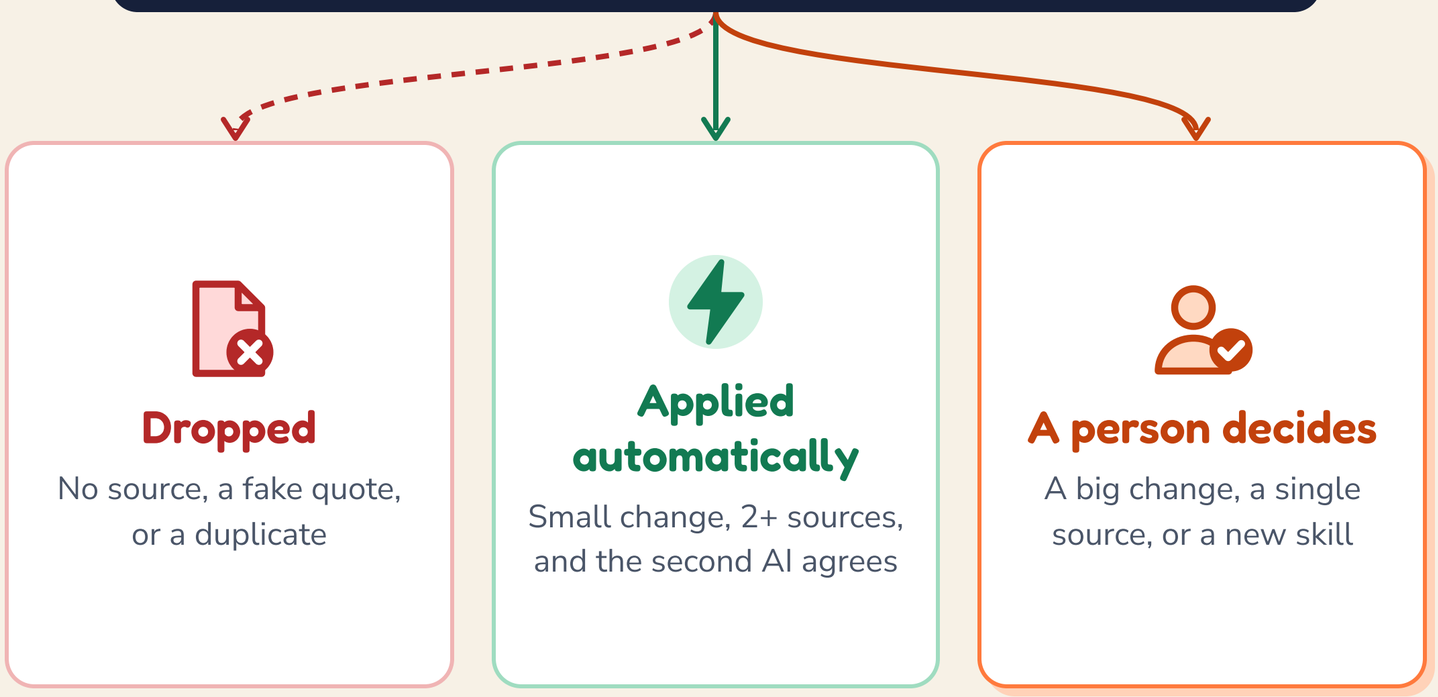
Adoption
Companies putting AI to work on those tasks

Demand
Employers needing more or fewer people with a skill

Live run, 23 Oct 2026. Each daily run covers 10 skills, so all 29 map skills every 3 days, plus 10 broad topics. Next: all 13,939.

Triage: a person approves every big change

Every finding is checked first
Trusted source, quote found word for word, then a second AI reviews it



Real signal, 22 Oct 2026

"...the Account Reconciliation Agent can help identify exceptions between subledgers and the general ledger and suggest how to resolve them..."

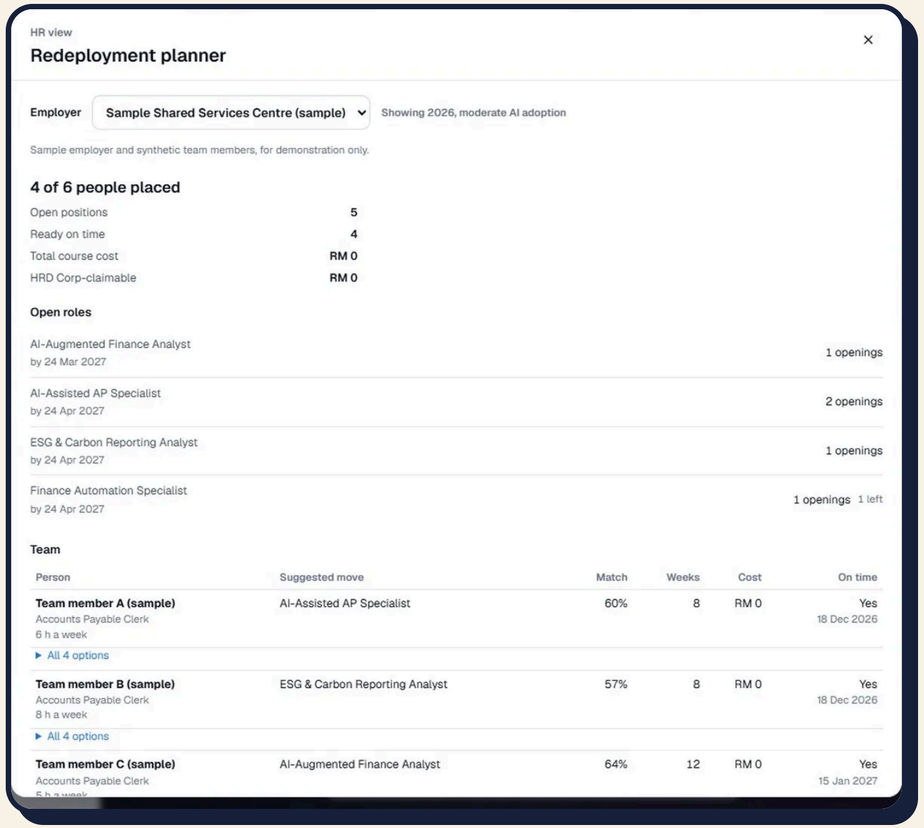
Microsoft Dynamics 365 blog, quote found on the page

Vendor reconciliation: AI exposure up, single source, so it went to a person

Approved → every route through reconciliation is re-planned

Every change is saved as a new version and can be undone. A new skill counts at half weight until a person confirms it.

Employers and advisors use the same map



HR view

Redeployment planner

Employer: Sample Shared Services Centre (sample) — Showing 2026, moderate AI adoption

Sample employer and synthetic team members, for demonstration only.

4 of 6 people placed

Open positions	5
Ready on time	4
Total course cost	RM 0
HRD Corp-claimable	RM 0

Open roles

AI-Augmented Finance Analyst by 24 Mar 2027	1 openings
AI-Assisted AP Specialist by 24 Apr 2027	2 openings
ESG & Carbon Reporting Analyst by 24 Apr 2027	1 openings
Finance Automation Specialist by 24 Apr 2027	1 openings 1 left

Team

Person	Suggested move	Match	Weeks	Cost	On time
Team member A (sample) Accounts Payable Clerk 6 h a week	AI-Assisted AP Specialist	60%	8	RM 0	Yes 18 Dec 2026
▸ All 4 options					
Team member B (sample) Accounts Payable Clerk 8 h a week	ESG & Carbon Reporting Analyst	57%	8	RM 0	Yes 18 Dec 2026
▸ All 4 options					
Team member C (sample) Accounts Payable Clerk	AI-Augmented Finance Analyst	64%	12	RM 0	Yes 15 Jan 2027

HR redeployment planner

Who moves where, by when, at what cost. Sample team: 4 of 6 placed, 4 ready on time, RM 0 in course fees.

Employer requirements

Employers add the skills they need; each worker’s match and route update.

Advisor approval

A plan becomes a learning plan only after an advisor signs off.

For every worker

Rank jobs by SDG goal, use English or Bahasa Melayu, delete your data anytime.

Live screenshot; sample employer and synthetic team, labelled in the app. The app suggests who moves where; HR makes the decision.

SDG mapping

Every skill and job links to a UN goal

One worker's route, goal by goal



Today's job

Accounts Payable Clerk

Highly exposed to AI; the aim is redeployment, not retrenchment



A skill on the route

GHG emissions accounting

Measures emissions so companies can cut them



Target job

ESG & Carbon Reporting Analyst

Measures company emissions; supports sustainability disclosure

Rank jobs by a goal

Pick a goal, such as climate action. Jobs that serve it gain 10 points on a 100-point score, and the app shows what the switch costs in match and weeks.



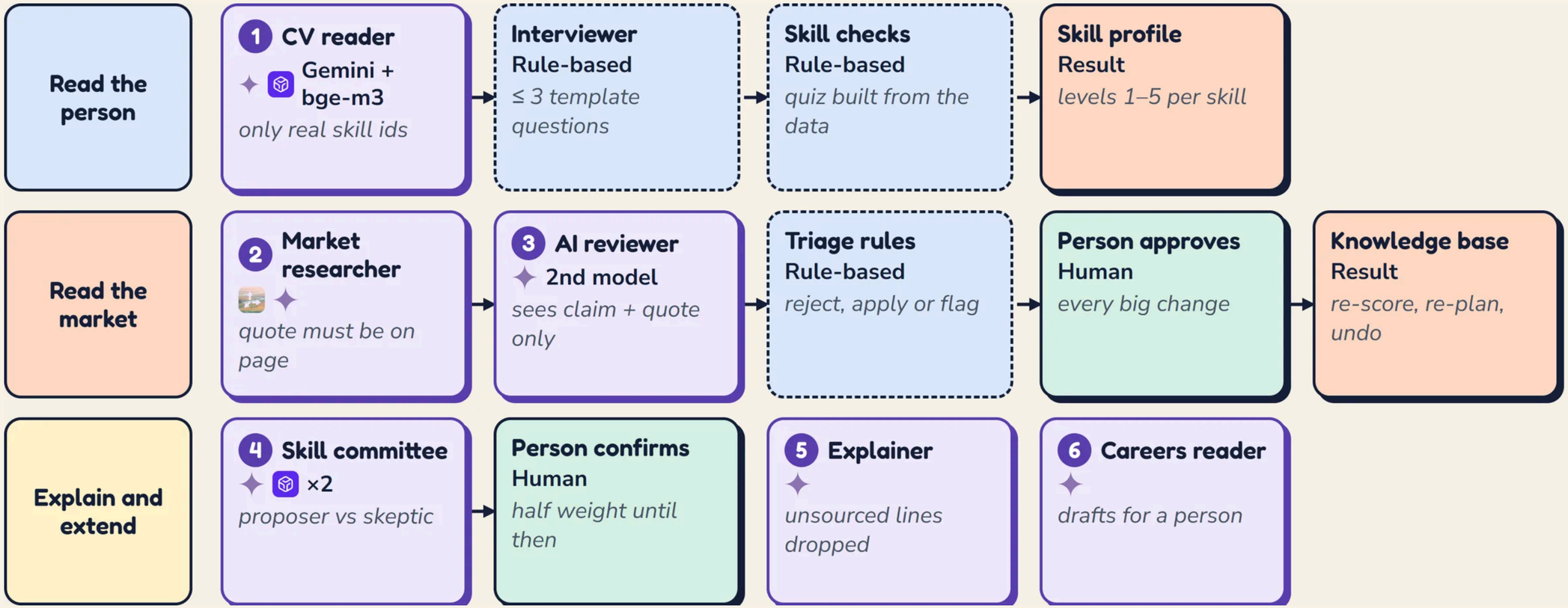


Reskilling itself serves SDG 4 (target 4.4). A test fails if any job or skill loses its tag.

Tags are our own labels on the sample catalogue, marked Sample in the app. Icons: United Nations SDGs.

Multi-agent design

Six AI agents, each checked by code or a person



Purple numbered boxes are AI agents, dashed blue boxes are fixed rules, green boxes are people. Code sets every number.

Proof

Every claim has a test behind it

CV reading

97%

current job right on 36 CVs it never saw, with a blind answer key

Privacy

99.4%

of 155 planted personal details removed before any AI reads the CV

Exposure model

r = 0.92

agreement with experts on 185 jobs it never saw in training

Independent check

r = 0.79

against the ILO's index on 2,956 jobs, which we never trained on

Market researcher

769

articles read in one 5.5-minute live run, 7 flagged for a person

Engine

419

automated tests on routes, the stress test and the HR view, all passing

Known gaps: 28% of full-ESCO skills shown are wrong (word match, wrong meaning); 1 of 155 personal details reached the AI.

Risks

Our hardest risk, and the fallbacks we built

HARDEST RISK — **A wrong AI signal reroutes thousands of plans**

OUR MITIGATION — Trusted sites only. The quote must match the page word for word. A second AI reviews it. A person approves anything that changes a route, and every change can be undone.

RISK — **Gemini is down or out of quota**

FALLBACK — CV reading drops to keyword matching, labelled “basic”. The researcher replays its last run, labelled “replayed”.

RISK — **RunPod is cold or down**

FALLBACK — Embeddings switch to a local bge-m3 server or Gemini, then to keyword matching.

RISK — **Our AI-exposure scores are off**

FALLBACK — The model’s measured error is built into the 1,000-future stress test, and four sources are shown side by side.

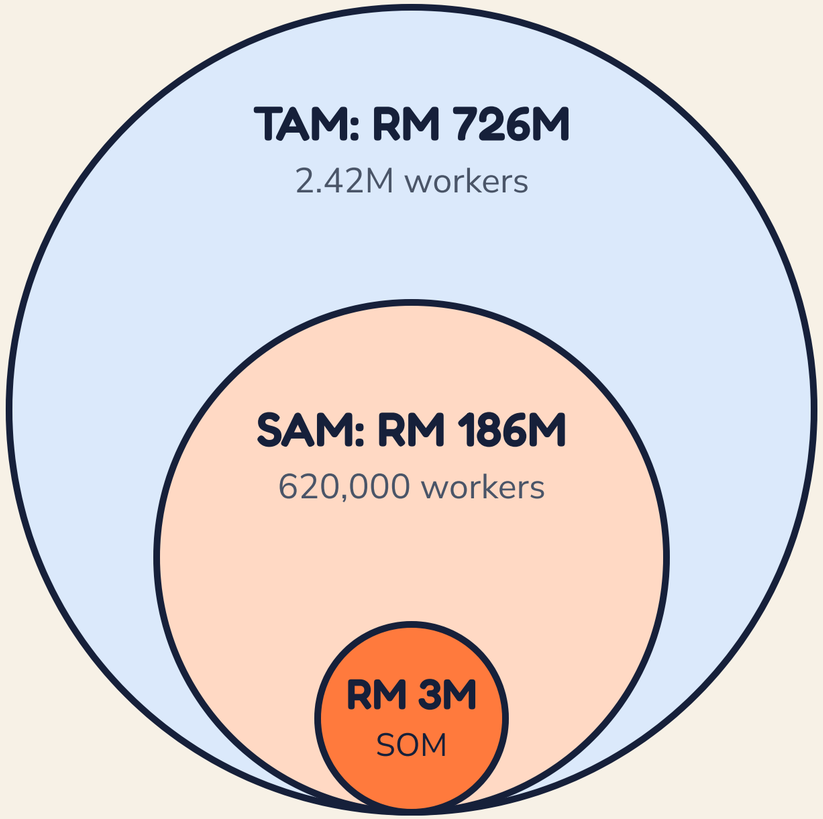
RISK — **Sample data taken as real**

FALLBACK — Every sample value is labelled Sample in the app; “check any job” runs on real ESCO data.

Every fallback is built, and the screen says when one is running.

Market size

RM 726M a year in Malaysia



TAM: workers who must move or upskill
620k high-impact and 1.8M medium-impact jobs in 10 sectors

SAM: jobs at high risk
Highly impacted roles that merge or disappear within 3–5 years

SOM: 10,000 seats by year 3
20 employers × 500 staff, starting in Penang and KL shared-services centres (749 GBS firms)

Sources: TalentCorp impact study 2024 and GBS Malaysia Mid-Term Report 2025. All at RM 300 per worker per year, our pilot price.

Business model

Free for workers, paid by employers

Why employers pay: one clerk, 11 years' service

Statutory severance

RM 22,800

HalaTuju Enterprise for six months

RM 150

One redeployment pays for about 150 workers.

Subscription models

Enterprise — Main revenue

RM 25 per worker a month, paid by the employer

Redeploy staff to internal roles, or help leavers find work. HR view, and staff opt in.

Personal

Free

Map, routes and stress test. Private to the worker.

Personal Plus

RM 9.90 a month

Small monthly fee for certificates and advisor sessions.

Public partners

Annual licence

PERKESO, TalentCorp and universities license it.

Severance: 20 days' wages per year of service at the DOSM 2025 clerical median, RM 2,693. Prices are our pilot assumptions.

Roadmap

From 29 skills to 13,939, then employers

Now: Oct 2026

Live app: 29 skills tracked daily, any of 3,039 jobs, 1,000-future stress test

Done: live demo, 36-CV blind test

Finals

Track all 13,939 skills. Import HR data from employers.

Target: letter of intent from 1 pilot employer

Pilot: H1 2027

Penang, home to about 90 shared-services companies such as **AMD**. First letters go to **InvestPenang** and **PSDC**.

Target: 2–3 centres, about 1,500 staff, redeployment rate measured

Scale: 2027–28

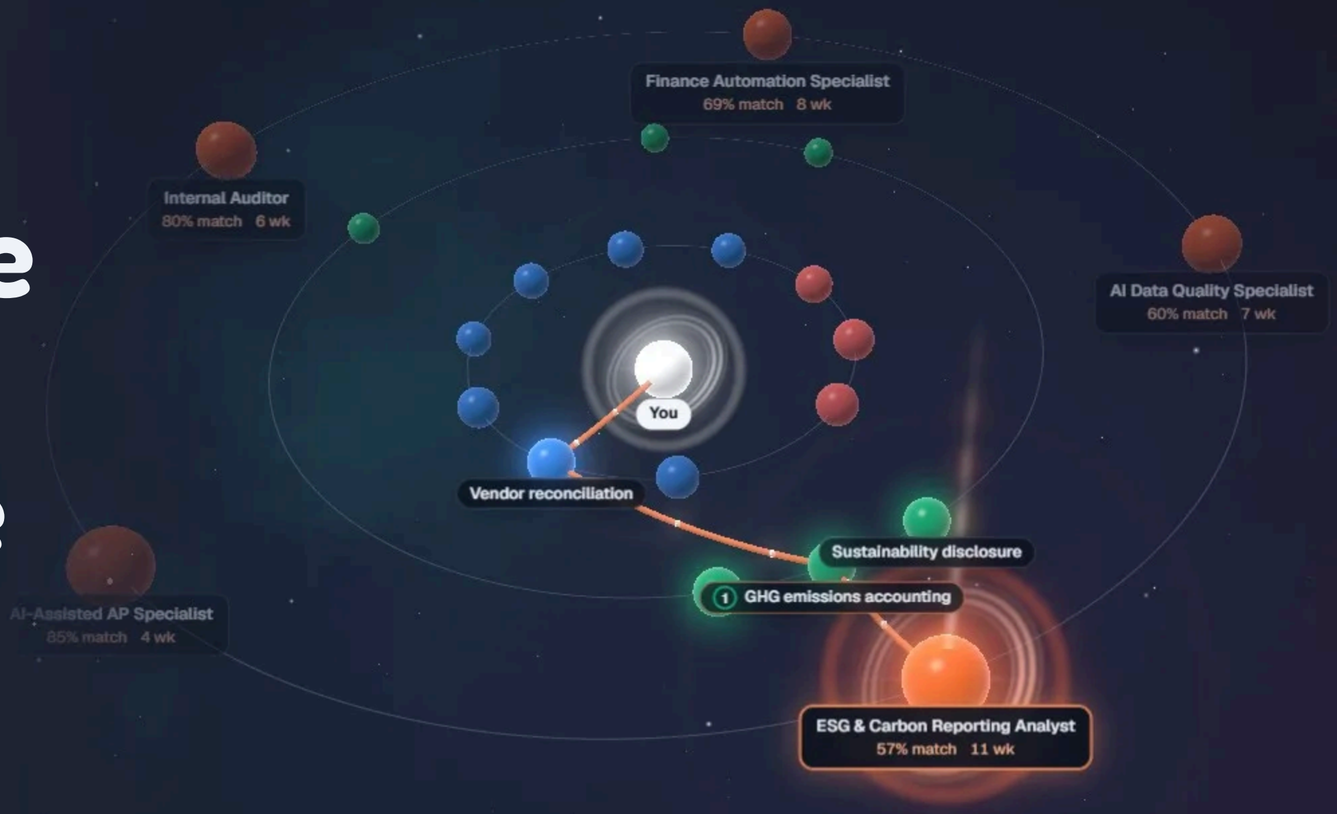
20 employers and universities. Licensed Malaysian job-ad data. More sectors.

Target: 10,000 seats, RM 3M a year

InvestPenang and PSDC are partners behind the Penang GBS Industry Academy. Not approached yet; AMD is an example, not a partner.

Thank you

Your career, rerouted the moment AI changes the road.



HalaTuju

Live demo: halatuju.replit.app

Code: github.com/shenenleow10521/SdcHackathon

Team Dsaise

Appendix: CV reading

36 unseen CVs, three kinds, scored blind

Kind	CVs	Job right	Found	Correct	Privacy
Clean	18	100%	71%	72%	100%
LinkedIn	12	92%	74%	65%	100%
Messy	6	100%	68%	77%	98%
All	36	97%	71%	72%	99.4%

On the app’s 29 map skills the same CVs score 94% found and 69% correct.

Known gaps

- Word match, wrong meaning: “payment proposal” read as *process payments*, which ESCO means as taking payments at a till
- “SQL Accounting”, a software package, read as the SQL language in 5 CVs
- 1 of 155 personal details reached the AI, on a two-column CV

Found and correct are for the 13,939 ESCO skills. 50 synthetic CVs in all; 14 used for tuning are left out here. Median 12.9 s per CV.

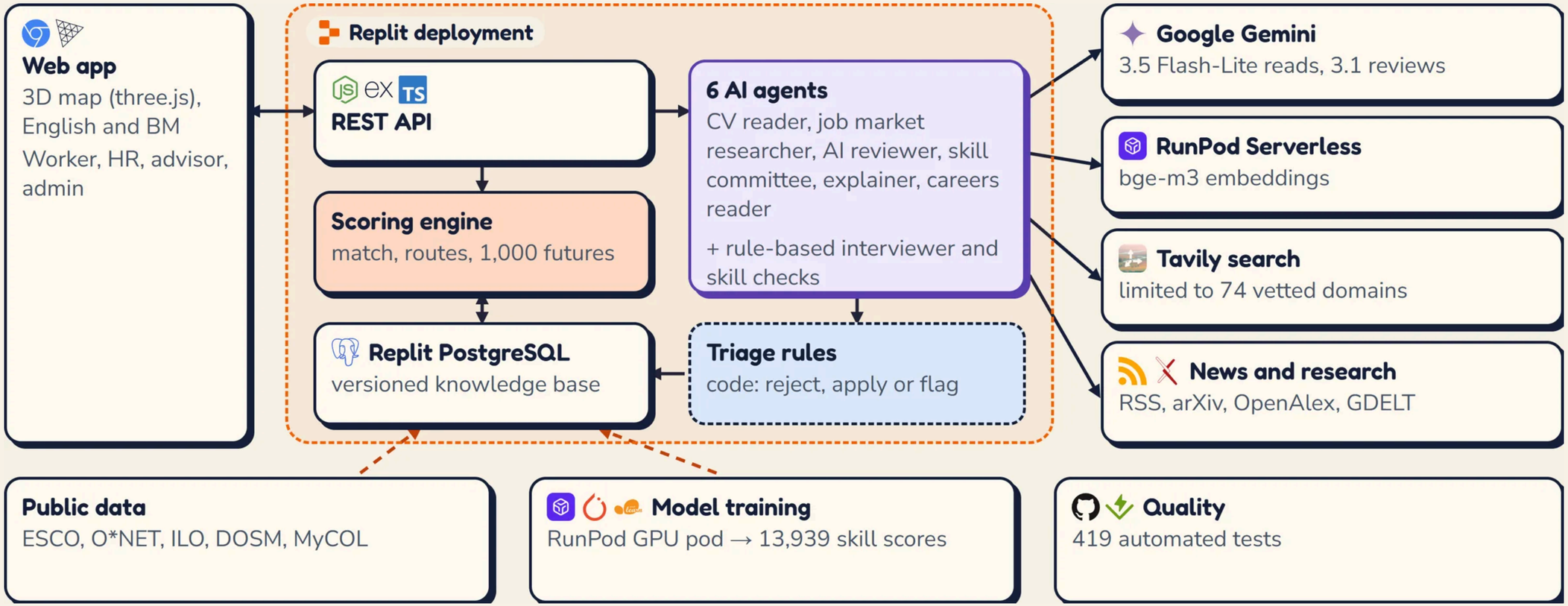
Appendix: what sets it apart

Five things you can check in the live app

What HalaTuju does	Evidence
A route to one specific job, in weeks	11 weeks at 6 h/week, every step linked to a real course
Every plan stress-tested	1,000 simulated futures (Monte Carlo); best route works in 100%
Reroutes from verified evidence	Quote found on the source page; a person approves
A measured exposure model	r = 0.92 on 185 held-out jobs; r = 0.79 vs the ILO index
Malaysian data on any job	DOSM wages and employment, TalentCorp critical-jobs list

Appendix: architecture

One Replit app, four outside services



AI reads text; plain code calculates every number. Dashed lines run once, offline: loading public data and training the model.

Appendix: how it is built

The stack, layer by layer

Data

ESCO v1.2.1
3,039 jobs and 13,939 skills

O*NET with GPTs-are-GPTs
16,411 expert-rated tasks

ILO WP140
AI exposure for 2,956 jobs

DOSM and TalentCorp MyCOL
Malaysian wages, critical jobs

AI

Gemini 3.5 Flash-Lite
reads CVs and news

Gemini 3.1 Flash-Lite
reviews each claim

bge-m3 on RunPod Serverless
matches skills and CVs

Exposure model on a RunPod GPU pod
PyTorch and scikit-learn

App

Replit
hosting and deployment

Node.js, Express, TypeScript
API and scoring engine

Replit PostgreSQL
versioned knowledge base

three.js
3D map, in English and Bahasa Melayu

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419 automated tests pass on GitHub with Vitest. Open source under Apache 2.0.

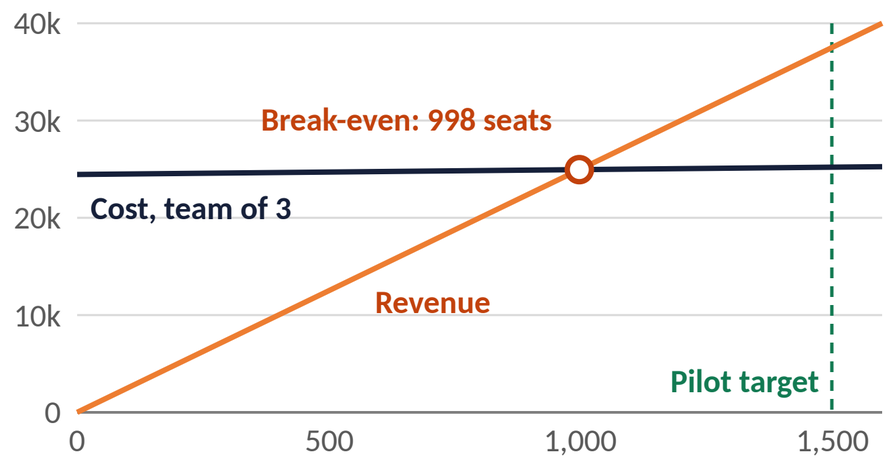
Appendix: costs and break-even

Breaks even at about 1,000 workers

Cost model, RM a month

	A	B	C
1	**Item**	**Value**	**Formula / basis**
2	Price per seat a month (RM)	25.00	*input*
3	Cost to serve per seat (RM)	0.50	*input, allowance*
4	Contribution per seat (RM)	24.50	=B2-B3
5	Read one CV (RM)	0.03	*measured*
6	One market check (RM)	1.05	*measured*
7	Daily checks a month (RM)	31.50	=B6*30
8	Hosting and database (RM)	420.00	*input, USD 100*
9	Team of 3 (RM)	24,000.00	*input, 3 x 8,000*
10	**Fixed cost a month (RM)**	**24,451.50**	=B7+B8+B9
11	**Break-even seats**	**998**	=B10/B4
12	Pilot target seats	1,500	*input*
13	**Profit a month at pilot (RM)**	**12,298.50**	=B12*B4-B10

Revenue and cost, RM a month, by seats



Blue = input (assumption), green = measured 24 Oct 2026, black = formula. USD 1 = RM 4.20.

Appendix: sources

Every number on these slides

- ESCO v1.2.1, European Commission: occupations and skills
- ILO Working Paper 140 (2025): generative AI exposure
- Eloundou et al., “GPTs are GPTs”, Science 384 (2024)
- Anthropic Economic Index, Massenkoff & McCrory (2026)
- O*NET task statements and ratings
- TalentCorp & ILMIA, MyCOL 2024/2025
- DOSM via OpenDOSM: wages by occupation, 2021–2025
- TalentCorp AI, digital and green economy impact study (Nov 2024)
- DOSM Labour Force Survey, Sep 2025: employment by occupation
- GBS Malaysia Strategy Mid-Term Report 2022–2027 (2025)
- HRD Corp levy collection 2024 (Ministry of Human Resources)
- Employment (Termination and Lay-Off Benefits) Regulations 1980
- Microsoft Dynamics 365 blog, 22 Oct 2026 (market researcher signal)
- HalaTuju measurements, 23–24 Oct 2026: CV test, engine, tests